CS342 · VISUAL PROGRAMMING

Midterm 461 Practice

Time Converter

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01 Questions

Problem GUI · 20 PTS

Our goal is to create a GUI that can convert a value in seconds into hours, minutes, and seconds.

a) Main Scenario: The user will enter a value between 0 and 86,400 seconds. When the 'Convert' button is pressed, the system will calculate the equivalent time in hours, minutes, and seconds. The input is represented by 's', and the outputs are 'ho' (hours), 'min' (minutes), and 'sec' (seconds). The formula used is: s = 3600 * ho + 60 * min + sec.

b) Exception Scenarios:
(1) EmptyFieldException: thrown when the input JTextField is left blank. Dialog "Empty Input": "Enter the seconds Value to convert".
(2) NumberFormatException: thrown when the input cannot be parsed as a number. Dialog "Error: Input Format": "Enter the seconds as a numerical Value".
(3) InvalidSecondsException: thrown when the input value falls outside the allowed range. Dialog "Error: Invalid Input Value": "Enter the seconds Value in the range [0..86400]."

Note: all the exception classes are already defined.

Example: Input 12,500 seconds → 3 hours, 28 minutes, and 20 seconds (3 * 3600 + 28 * 60 + 20 = 12,500).



Steps TO IMPLEMENT

(1) Create a class called TimeConverter, defined as the main GUI window. (5 Marks)
(2) Define all GUI components (buttons, text fields, labels) as attributes of the class. (3 Marks)
(3) Handle exceptions with JOptionPane.showMessageDialog. (6 Marks)
(4) Use inner classes BConvert and BCancel to handle events (BConvert for conversion, BCancel for clearing). (6 Marks)
(5) Set the GUI size to 450x100 with setSize(450, 100) in the constructor.